Practical No. 4 - Math and Random

localhost:8888/notebooks/Practical%20No.%204%20-%20Math%20and%20Random%20Modules.ipynb

jupyter Practical No. 4 - Math and Random Modules Last Checkpoint: 5 minutes ago

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Code

In [1]:
```
# Practical No. 4
# Aim: Demonstrate Python math and random modules
import math
import random

def basic_math_operations():
    num1 = 16
    num2 = 4
    print("Basic Math Operations:")
    print(f"Square root of {num1}: {math.sqrt(num1)}")
    print(f"{num1} raised to the power of {num2}: {math.pow(num1, num2)}")
    print(f"Sine of {num1} degrees: {math.sin(math.radians(num1)):.4f}")
    print(f"Cosine of {num2} degrees: {math.cos(math.radians(num2)):.4f}")
    print(f"Factorial of {num2}: {math.factorial(num2)}")

def random_number_operations():
    print("\nRandom Number Operations:")
    print("Random float between 0 and 1:"; random.random())
    print("Random integer between 10 and 50:"; random.randint(10, 50))
    print("Random choice from a list:"; random.choice([2, 4, 6, 8, 10]))

    random_list = [1, 2, 3, 4, 5]
    random.shuffle(random_list)
    print("Shuffled list:"; random_list)
    print("Random sample of 3 elements from the list:", random.sample(random_list, 3))

if __name__ == "__main__";
    basic_math_operations()
    random_number_operations()
```

```
Basic Math Operations:
Square root of 16: 4.0
16 raised to the power of 4: 65536.0
Sine of 16 degrees: 0.2756
Cosine of 4 degrees: 0.9976
Factorial of 4: 24

Random Number Operations:
Random float between 0 and 1: 0.6382
Random integer between 10 and 50: 37
Random choice from a list: 8
Shuffled list: [3, 1, 5, 2, 4]
Random sample of 3 elements from the list: [2, 4, 1]
```